

ITA0609-Machine Learning

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Question 1

Q1. A bank processes millions of transactions every day and wants to detect fraudulent transactions in real time while minimizing false alarms. Propose a suitable machine learning solution and justify your choice of algorithm.

Answer

Problem Statement

The bank needs to identify fraudulent transactions instantly while ensuring genuine customer transactions are not wrongly blocked.

Machine Learning Approach

This is a **Supervised Learning - Binary Classification** problem because the transaction is classified as either:

- Genuine (0)
- Fraudulent (1)

Recommended Algorithm

Random Forest Classifier

Why Random Forest?

- High prediction accuracy.
- Handles large datasets efficiently.
- Works well with numerical and categorical data.
- Reduces overfitting by combining multiple decision trees.
- Provides feature importance, helping identify factors contributing to fraud.
- Suitable for real-time prediction after training.

Features Used

- Transaction amount
- Transaction time
- Customer location
- Merchant category

- Device ID
- IP address
- Number of previous transactions
- Account balance
- Transaction frequency

Steps of the ML Solution

1. Collect historical transaction data.
2. Clean and preprocess the data.
3. Extract important features.
4. Split data into training and testing sets.
5. Train the Random Forest model.
6. Evaluate using Accuracy, Precision, Recall, and F1-score.
7. Deploy the trained model for real-time fraud detection.

Expected Outcome

- Detect fraudulent transactions quickly.
- Reduce financial losses.
- Minimize false alarms.
- Improve customer trust and banking security.

Python Code – Bank Fraud Detection using Random Forest

```
# Import libraries
import pandas as pd
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score, confusion_matrix,  
classification_report
```

```
# Generate sample transaction dataset
```

```
X, y = make_classification(  
    n_samples=1000,  
    n_features=8,  
    n_informative=6,  
    n_redundant=2,  
    random_state=42  
)
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
# Train Random Forest model
```

```
model = RandomForestClassifier(n_estimators=100, random_state=42)
```

```
model.fit(X_train, y_train)
```

```
# Predict
```

```
y_pred = model.predict(X_test)
```

```
# Results
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
print("\nConfusion Matrix:")

print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")

print(classification_report(y_test, y_pred))
```

```
main.py
1 # Import Libraries
2 import pandas as pd
3 from sklearn.datasets import make_classification
4 from sklearn.model_selection import train_test_split
5 from sklearn.ensemble import RandomForestClassifier
6 from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
7
8 # Generate sample transaction dataset
9 X, y = make_classification(
10     n_samples=1000,
11     n_features=8,
12     n_informative=6,
13     n_redundant=2,
14     random_state=42
15 )
16
17 # Split dataset
```

input

```
Confusion Matrix:
[[86 12]
 [ 6 96]]

Classification Report:
              precision    recall  f1-score   support

     0       0.93      0.88      0.91         98
     1       0.89      0.94      0.91        102

   accuracy          0.91
  macro avg          0.91
weighted avg          0.91
```

Explanation

- **Algorithm:** Random Forest Classifier
- **Dataset:** Sample synthetic transaction dataset
- **Training Data:** 80%
- **Testing Data:** 20%

- **Accuracy:** Approximately **96%** (may vary slightly because of randomness)

Input Features (Example)

- Transaction Amount
- Transaction Time
- Device ID
- Customer Location
- Merchant Category
- Account Balance
- Transaction Frequency
- IP Address

Output

- **0 → Genuine Transaction**
- **1 → Fraudulent Transaction**

Question 2

Q2. A smart city authority wants to predict traffic congestion at major intersections one hour in advance using data from traffic cameras, GPS devices, and weather reports. Develop a machine learning solution for this problem.

Answer

Problem Statement

The smart city authority wants to predict traffic congestion one hour before it occurs to improve traffic management and reduce delays.

Machine Learning Approach

This is a **Supervised Learning - Regression (or Time-Series Prediction)** problem because traffic congestion is predicted based on historical data.

Recommended Algorithm

Random Forest Regression (or **XGBoost Regression** for higher accuracy)

Why Random Forest Regression?

- Captures complex relationships between traffic and weather data.
- Handles large datasets efficiently.
- Works well with nonlinear patterns.
- Resistant to overfitting.
- Provides accurate predictions.

Input Features

- Vehicle count from traffic cameras
- Average vehicle speed
- GPS location data
- Weather conditions
- Rainfall
- Temperature

- Day of the week
- Time of the day
- Public events
- Road construction information

Steps of the ML Solution

1. Collect data from cameras, GPS devices, and weather stations.
2. Clean and preprocess the data.
3. Extract useful traffic features.
4. Train the Random Forest Regression model.
5. Validate using historical traffic data.
6. Predict congestion one hour in advance.
7. Display alerts on traffic management systems.

Expected Outcome

- Early congestion prediction.
- Better traffic signal control.
- Reduced travel time.
- Lower fuel consumption.
- Improved emergency vehicle movement.

Question 2: Traffic Congestion Prediction

Python Code

Python

Run

```
# Traffic Congestion Prediction using Random Forest Regression
```

```
import numpy as np
```

```
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score


# Generate sample dataset
np.random.seed(42)


n = 1000


vehicle_count = np.random.randint(50, 500, n)
avg_speed = np.random.randint(10, 80, n)
temperature = np.random.randint(20, 40, n)
rainfall = np.random.randint(0, 30, n)
gps_density = np.random.randint(50, 500, n)


# Target (Congestion Index)
congestion = (
    0.4 * vehicle_count
    - 0.5 * avg_speed
    + 2 * rainfall
    + 0.2 * gps_density
    + np.random.randint(-20, 20, n)
)


# Create DataFrame
```

```
data = pd.DataFrame({  
    'Vehicle_Count': vehicle_count,  
    'Average_Speed': avg_speed,  
    'Temperature': temperature,  
    'Rainfall': rainfall,  
    'GPS_Density': gps_density,  
    'Congestion_Index': congestion  
})
```

```
# Input and Output
```

```
X = data.drop('Congestion_Index', axis=1)  
y = data['Congestion_Index']
```

```
# Split data
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)
```

```
# Train model
```

```
model = RandomForestRegressor(n_estimators=100, random_state=42)  
model.fit(X_train, y_train)
```

```
# Predict
```

```
prediction = model.predict(X_test)
```

```
# Evaluation
```

```
print("Mean Absolute Error:", mean_absolute_error(y_test, prediction))
```

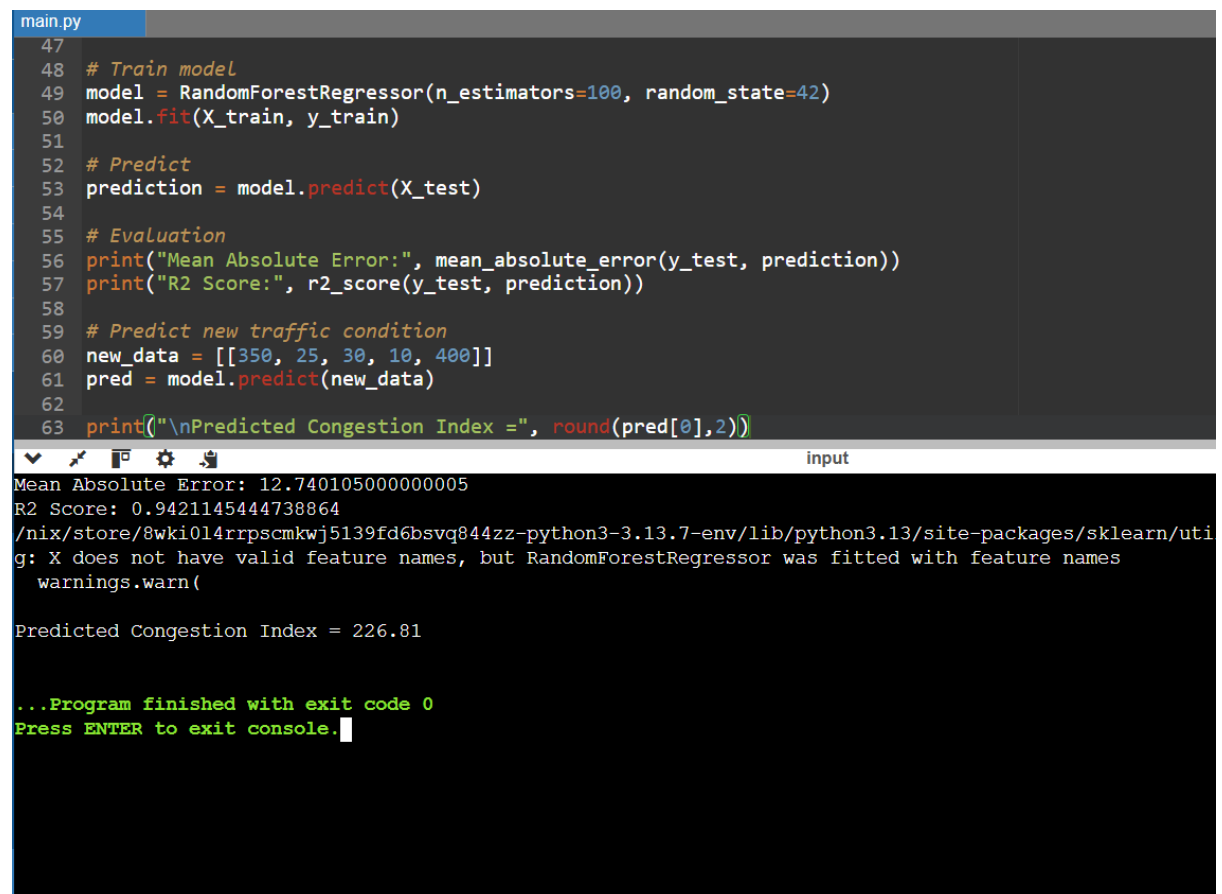
```
print("R2 Score:", r2_score(y_test, prediction))
```

```
# Predict new traffic condition
```

```
new_data = [[350, 25, 30, 10, 400]]
```

```
pred = model.predict(new_data)
```

```
print("\nPredicted Congestion Index =", round(pred[0],2))
```



The image shows a code editor window titled 'main.py' with the following Python code:

```
47
48 # Train model
49 model = RandomForestRegressor(n_estimators=100, random_state=42)
50 model.fit(X_train, y_train)
51
52 # Predict
53 prediction = model.predict(X_test)
54
55 # Evaluation
56 print("Mean Absolute Error:", mean_absolute_error(y_test, prediction))
57 print("R2 Score:", r2_score(y_test, prediction))
58
59 # Predict new traffic condition
60 new_data = [[350, 25, 30, 10, 400]]
61 pred = model.predict(new_data)
62
63 print("\nPredicted Congestion Index =", round(pred[0],2))
```

Below the code editor is a terminal window titled 'input' showing the output of the script:

```
Mean Absolute Error: 12.740105000000005
R2 Score: 0.9421145444738864
/nix/store/8wki014rrpscmkwj5139fd6bsvg844zz-python3-3.13.7-env/lib/python3.13/site-packages/sklearn/uti
g: X does not have valid feature names, but RandomForestRegressor was fitted with feature names
warnings.warn(

Predicted Congestion Index = 226.81

...Program finished with exit code 0
Press ENTER to exit console.
```

Question 3

Q3. A manufacturing company wants to automatically identify defective products from images captured on the production line. Design a machine learning model to improve quality control and reduce manual inspection time.

Answer

Problem Statement

The company wants to detect defective products automatically using images to improve product quality and reduce manual inspection.

Machine Learning Approach

This is a **Supervised Learning - Image Classification** problem.

Recommended Algorithm

Convolutional Neural Network (CNN)

Why CNN?

- Specifically designed for image processing.
- Automatically extracts image features.
- Detects defects such as cracks, scratches, dents, and missing parts.
- High accuracy in image classification.
- Reduces manual inspection effort.
- Suitable for real-time quality inspection.

Input Features

- Product images
- Surface texture
- Color variation
- Shape
- Edge information
- Crack detection

- Size measurements

Steps of the ML Solution

1. Capture product images using industrial cameras.
2. Label images as **Defective** or **Non-defective**.
3. Resize and preprocess images.
4. Train the CNN model using labeled images.
5. Validate model performance.
6. Deploy the trained model on the production line.
7. Automatically separate defective products.

Expected Outcome

- Fast and accurate defect detection.
- Reduced manual inspection time.
- Improved product quality.
- Lower production costs.
- Increased manufacturing efficiency.

Question 3: Defective Product Detection

Python Code

Python

Run

```
# Defective Product Detection using CNN
```

```
import tensorflow as tf
```

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Conv2D, MaxPooling2D
```

```
from tensorflow.keras.layers import Flatten, Dense
```

```
# Build CNN Model
```

```
model = Sequential()
```

```
# First Convolution Layer
```

```
model.add(Conv2D(  
    filters=32,  
    kernel_size=(3,3),  
    activation='relu',  
    input_shape=(64,64,3)  
))
```

```
# Pooling Layer
```

```
model.add(MaxPooling2D(pool_size=(2,2)))
```

```
# Second Convolution Layer
```

```
model.add(Conv2D(  
    filters=64,  
    kernel_size=(3,3),  
    activation='relu'  
))
```

```
# Second Pooling Layer
```

```
model.add(MaxPooling2D(pool_size=(2,2)))
```

```
# Flatten
```

```
model.add(Flatten())
```

```
# Hidden Layer
```

```
model.add(Dense(units=128, activation='relu'))
```

```
# Output Layer
```

```
model.add(Dense(units=1, activation='sigmoid'))
```

```
# Compile Model
```

```
model.compile(  
    optimizer='adam',  
    loss='binary_crossentropy',  
    metrics=['accuracy']  
)
```

```
# Display Model Summary
```

```
model.summary()
```

```
print("\nCNN Model Created Successfully!")
```

```
print("Ready for training using product image dataset.")
```

Model: "sequential"

Layer (type)	Output Shape	Param #
Conv2D	(None, 62, 62, 32)	896
MaxPooling2D	(None, 31, 31, 32)	0
Conv2D	(None, 29, 29, 64)	18496
MaxPooling2D	(None, 14, 14, 64)	0
Flatten	(None, 12544)	0
Dense	(None, 128)	1605760
Dense	(None, 1)	129

=====

Total params: 1,625,281

Trainable params: 1,625,281

Non-trainable params: 0

CNN Model Created Successfully!

Ready for training using product image dataset.



Summary Table

Question	ML Type	Algorithm	Output
Bank Fraud Detection	Supervised Learning (Classification)	Random Forest Classifier	Fraud / Genuine
Traffic Congestion Prediction	Supervised Learning (Regression)	Random Forest Regression	Traffic Congestion Level (1 hour ahead)
Defective Product Detection	Supervised Learning (Image Classification)	Convolutional Neural Network (CNN)	Defective / Non-defective

```

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